

## Study Guide

*this is a place to start, it is not ALL inclusive*

*learn the big picture first; then dive into the details; seek understanding not memorization*

### L16: Metabolism

1. Differentiate digestion, catabolism, anabolism; know basic biochemistry of macromolecules & monomers
2. What is a nutrient? Know essential nutrients required by the body (not specific examples).
3. Describe the components of energy balance and what is required for someone to be in energy balance (no change in weight), and what happens in energy balance for someone to lose or gain weight.
4. Define adaptive thermogenesis.
5. Define basal metabolic rate and the state a person must be in to be burning fuel ONLY to sustain their basal metabolic rate.
6. Distinguish metabolic rate and basal metabolic rate. What factors contribute to metabolic rate?
7. Describe the regulation of food intake; include the 2 hormones involved.
8. Distinguish between visceral and subcutaneous fat; which promotes systemic inflammation?
9. **Potential short answer:** Describe the negative feedback loop for body temperature.
10. **Potential essay:** describe the relationship between ambient temperature, body temperature and metabolic rate as presented in the graphs from lecture. Graphs include **body temperature vs. environmental temperature** and **metabolic rate vs environmental temperature**.
11. **Potential short answer:** Describe the negative feedback loop involved in regulation of **basal metabolic rate**. (NOTE: you already know this from endocrinology. You will NOT have to know the diseases as we already went over these in a past lecture.)
12. What is meant by the terms absorptive & postabsorptive states.
13. Define these terms: glycogenesis, gluconeogenesis, glycolysis, glycogenolysis, lipolysis, lipogenesis, ketogenesis. In what body tissues do each of these occur?
14. **Potential essay:** Explain the roles of the liver, skeletal muscle and adipose tissue in blood fuel levels during fasted and feasted states. This is **short term**, not long-term regulation; this involves the pancreas and 2 hormones.
15. What type of events (not eating) require us to have mechanisms for **long term** regulation of blood glucose levels?
16. **Potential essay/short answer:** Describe the 2 branches of the "Stress Response". Which branch is primarily used to respond to short term acute stress that resolves quickly? Which branch is primarily used to respond to chronic stress that does not resolve quickly?

## L16: Metabolism – Diabetes

1. Distinguish between diabetes mellitus and diabetes insipidus and their treatments
2. **Potential short answer:** Distinguish between type I and type II diabetes.
3. Why is high blood glucose a health risk both in the short term and over time?
4. How does insulin binding to its receptor cause glucose to enter cells?
5. **Potential short answer:** List and explain the 3 symptoms of diabetes. Make sure to explain the physiology behind each resulting symptom.
6. **Potential short answer:** Explain how the consequences of uncontrolled blood sugar in a type I diabetic can cause death within hours.
7. What is hypoglycemia, what are symptoms, and how is it treated?